

Predicting Future Greenhouse Gas Emissions using Machine Learning

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Introduction

Climate change mitigation is one of the most pressing global challenges. Predicting greenhouse gas (GHG) emissions is crucial for informed policy-making and sustainable development. Our research leverages machine learning (ML) techniques to predict GHG emissions per capita for 175 countries using historical data from 2000 to 2015.

Research Question

How can machine learning models predict annual GHG emissions of a country using historical environmental and socioeconomic data?

Datasets & Features

Source: World Resources Institute, World Bank

Timeframe: 2000-2015

Features Used:

- Fossil Fuel Energy Consumption
- Renewable Energy Usage
- GDP per Capita
- Urban Population
- Access to Electricity
- Total Final Energy Consumption

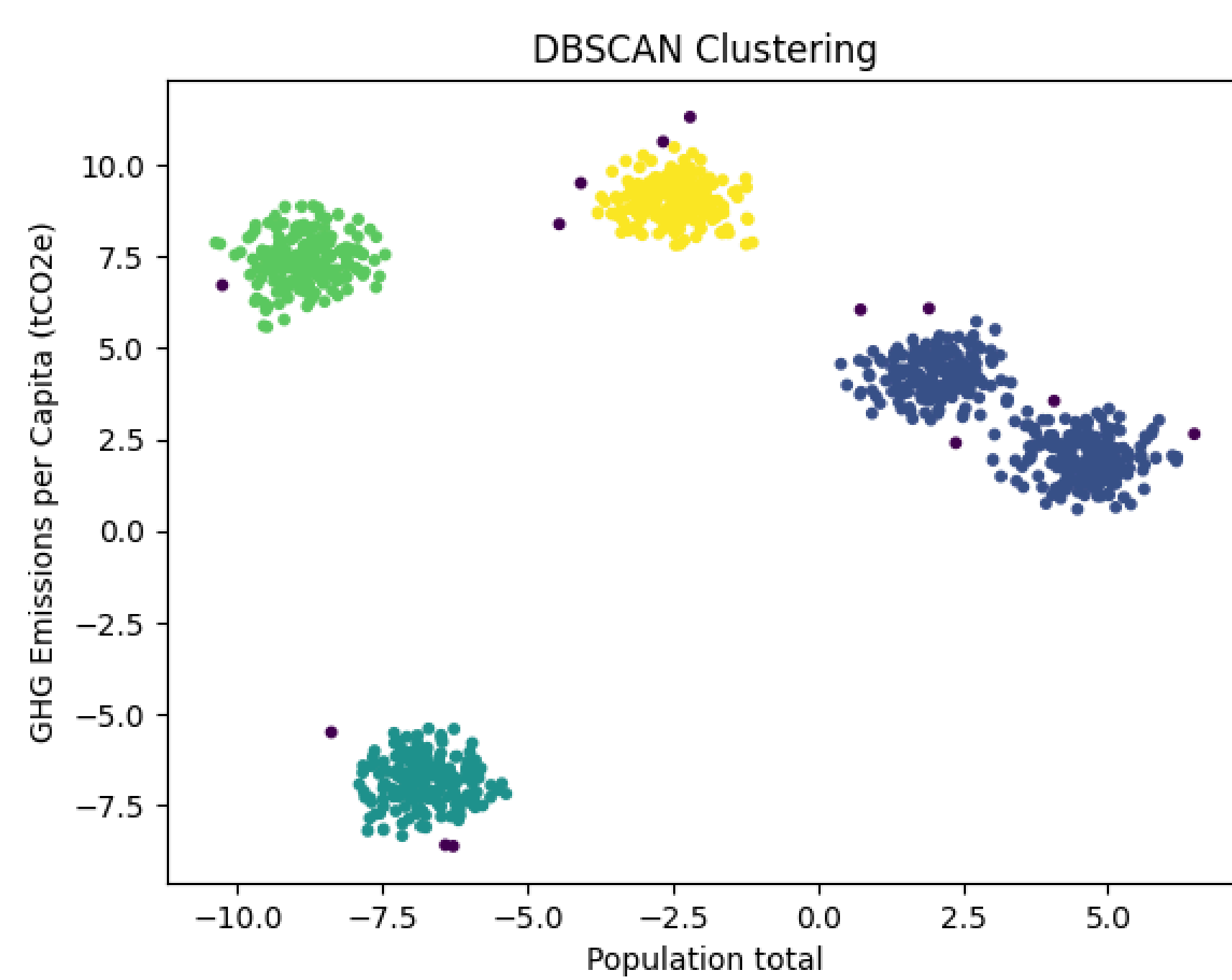


Figure 1. represents **DBSCAN Clustering** of GHG Emissions per Capita vs. Population Total. The clusters indicate distinct groups of countries based on their emissions and population sizes, with densely populated areas forming core clusters (colored points) and outliers (purple points) identified as noise.

Neural Networks

```
# Split dataset into features and target variable
X_nn = df_new.drop(columns=['GHG Emission Class', 'CountryID', 'Country', 'Year', 'GHG Emissions per Capita (tCO2e)'])
y_nn = df_new['GHG Emissions per Capita (tCO2e)'] # Target variable

# Splitting the dataset into training and testing sets
X_train_nn, X_test_nn, y_train_nn, y_test_nn = train_test_split(X_nn, y_nn, test_size=0.2, random_state=42)

# Feature scaling
scaler = MinMaxScaler()

# Fit scaler to training data and transform both training and testing data
X_train_nn_scaled = scaler.fit_transform(X_train_nn)
X_test_nn_scaled = scaler.transform(X_test_nn)
```

	Agricultural land (% of land area)	Population, total	Total electricity output per capita (kWh)	Total final energy consumption per Capita (TFEC) (MJ)	GHG Emissions per Capita (tCO2e)
count	2592.000000	2.800000e+03	2576.000000	2608.000000	2800.000000
mean	39.905523	3.644690e+07	3511.743131	51224.460894	4.812985
std	21.584310	1.384211e+08	5521.685286	57010.209942	6.681345
min	0.448718	9.609000e+03	0.000000	0.000000	0.019519
25%	21.712776	1.899871e+06	284.113336	14003.981099	0.593190
50%	41.084986	7.557865e+06	1575.915850	28953.459523	2.357686
75%	57.436346	2.302792e+07	4623.777843	68781.725064	6.610386
max	85.487373	1.379860e+09	56826.322869	365436.142945	67.498391

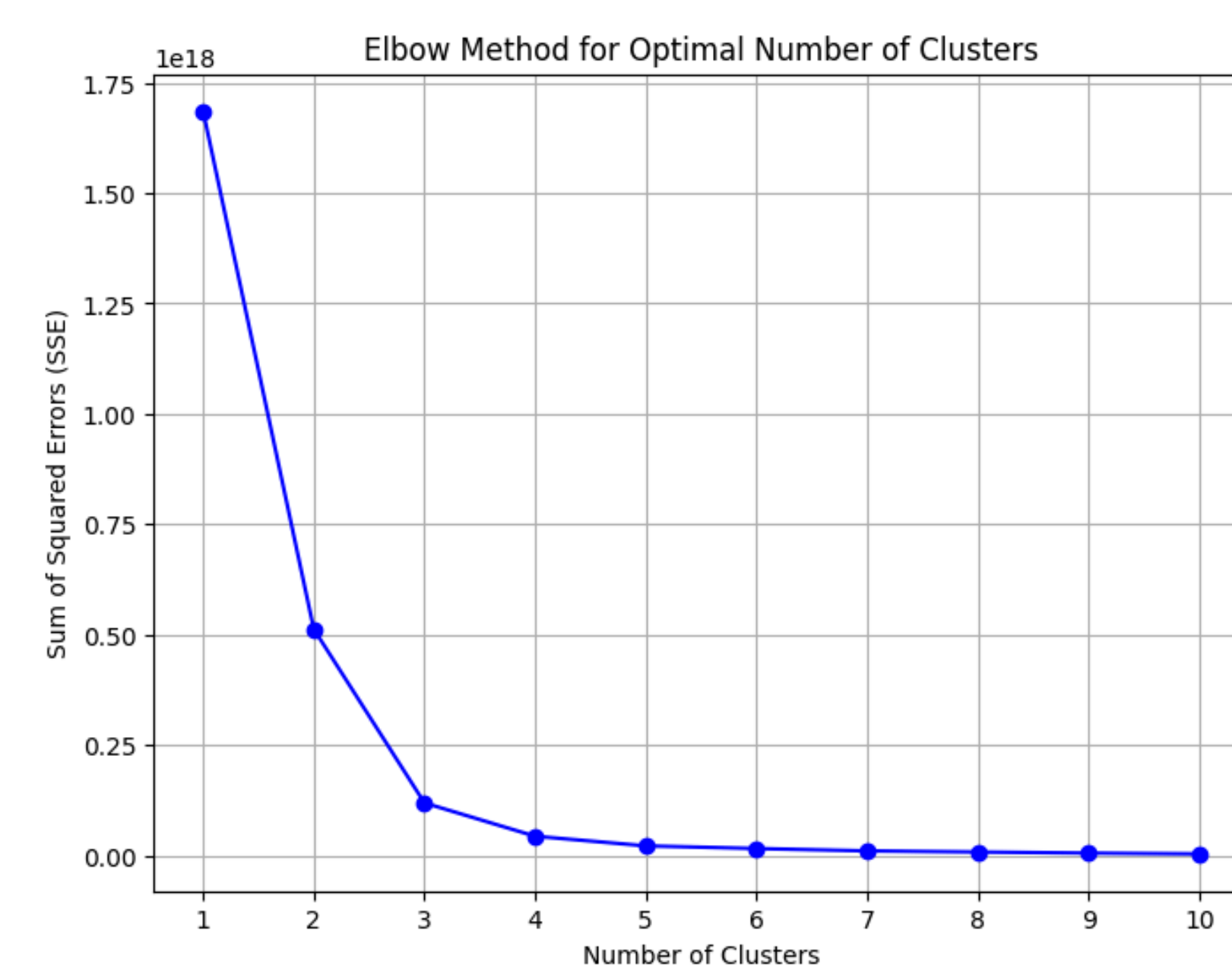


Figure 2. represents the **Elbow Method** for determining the optimal number of clusters in a dataset using Sum of Squared Errors (SSE). The "elbow point" around (k=3) indicates the ideal number of clusters.

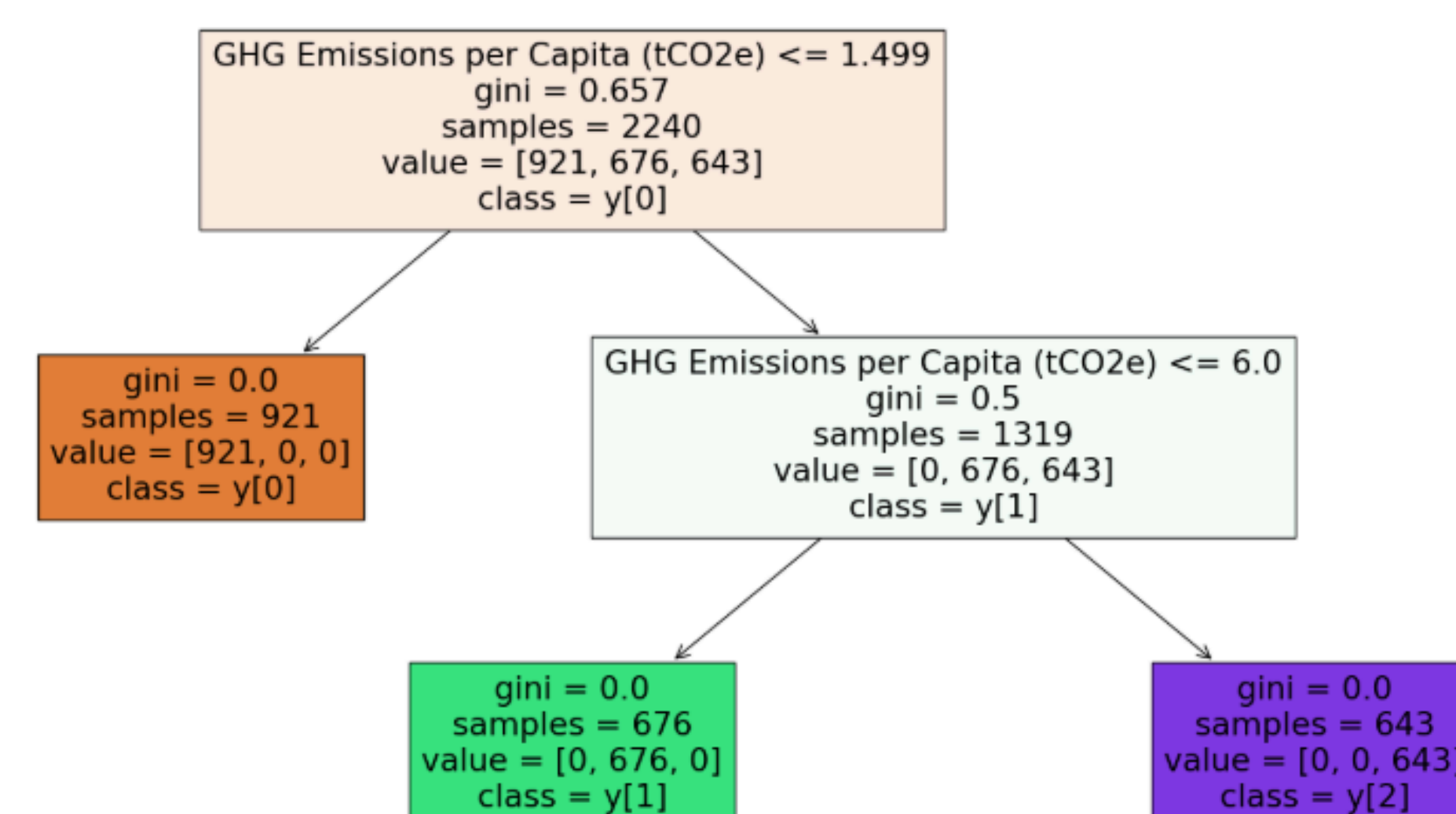


Fig 4. This is a **Decision Tree Classification** Output for predicting GHG Emissions per Capita (tCO2e) based on historical environmental and socioeconomic indicators.

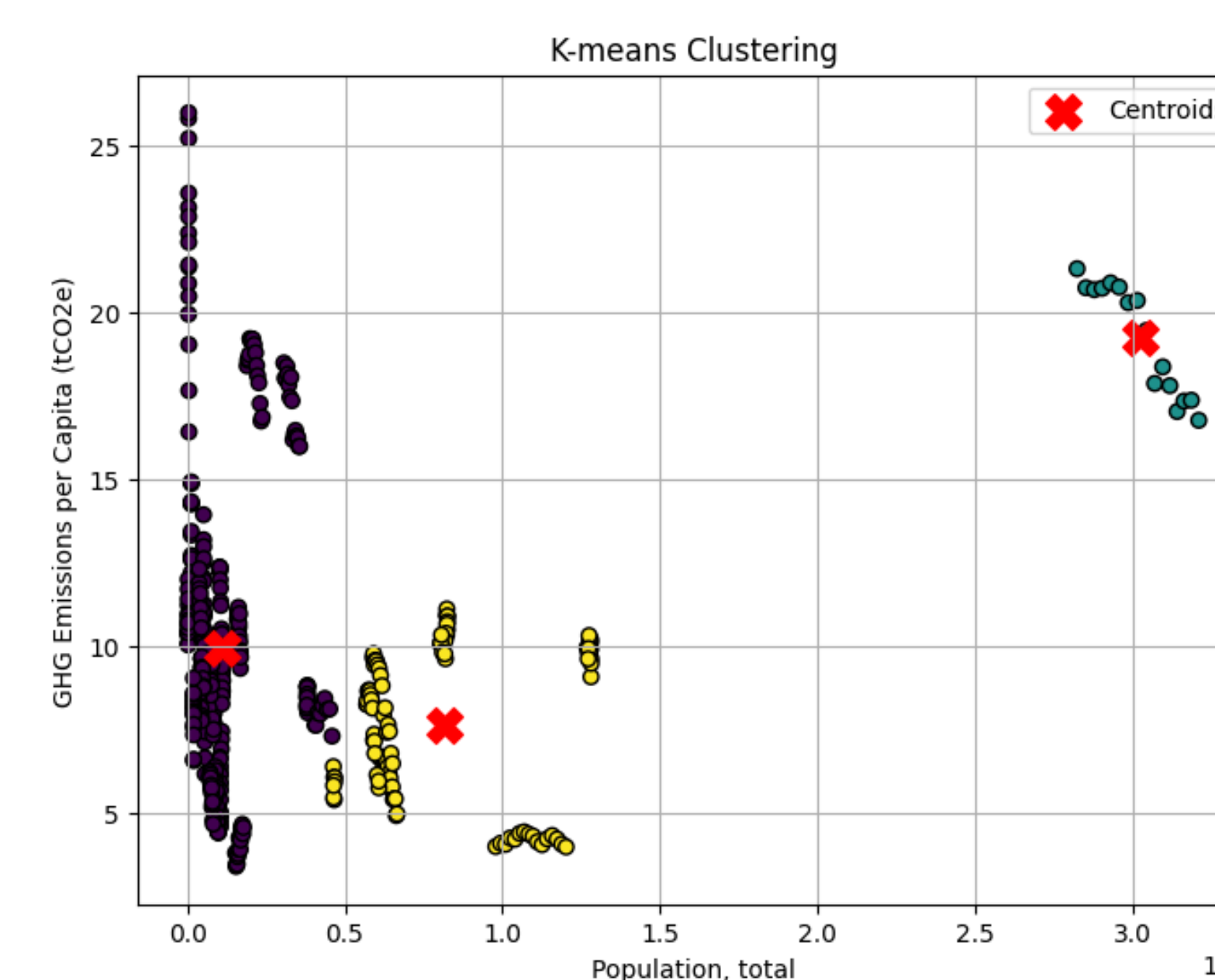


Figure 3. represents K-Means Clustering of GHG Emissions per Capita vs. Population Total, where different colors indicate distinct clusters, and red crosses mark the centroids of each cluster.

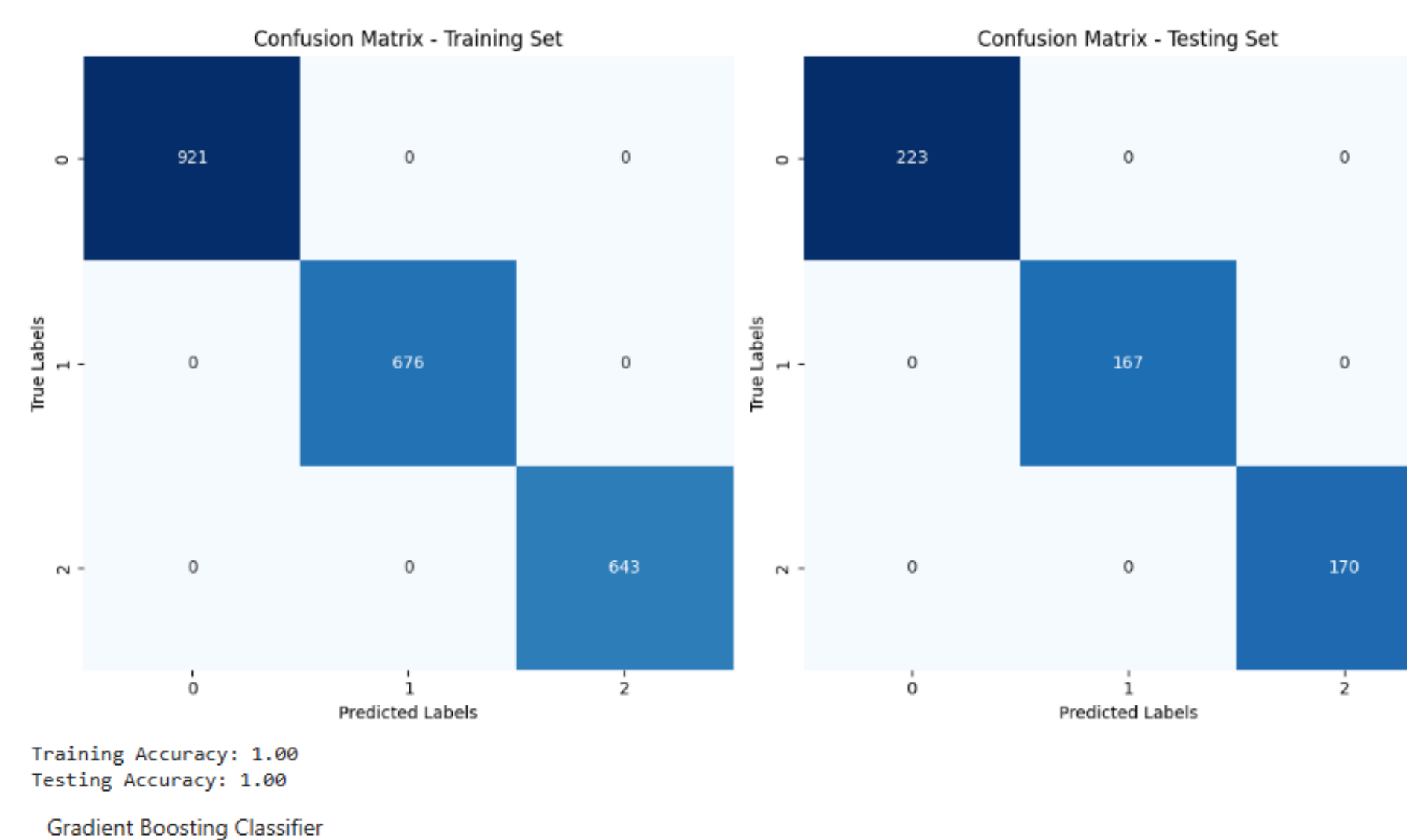


Figure 5. displays **Confusion Matrices** for the training and testing sets of a **Gradient Boosting Classifier** used to predict GHG emission levels. Each matrix shows perfect classification, with 100% accuracy for both sets.

Methodology

We first compiled a dataset containing **16 predictive features** across **175 countries**, applying data cleaning and normalization for consistency. Using this dataset, we tested multiple ML models, including Linear Regression, Logistic Regression, Decision Trees, Random Forests, and Support Vector Regression (SVR). The SVR model with a Radial Basis Function (RBF) kernel performed best among traditional approaches, showing the lowest RMSE. To further enhance predictive accuracy, we explored advanced machine learning techniques in the final phase, implementing neural networks with multiple hidden layers and Scaled Exponential Linear Unit (SELU) activation, which achieved the highest accuracy (**RMSE = 2.54, Correlation = 89.65%**). Additionally, clustering techniques such as K-Means and Hierarchical Clustering revealed distinct patterns in GHG emissions based on socioeconomic indicators.

Results

The key findings of this research highlight the significant role of **Total Electricity Output per Capita (kWh)** and **GDP per Capita** as the most predictive features for greenhouse gas (GHG) emissions, which aligns with existing climate and economic studies. Higher energy consumption directly correlates with increased emissions, while GDP per capita serves as an indicator of industrialization, economic activities, and energy usage patterns.

Conclusion

Machine learning effectively predicts GHG emissions, with Total Electric Output and GDP per Capita as key predictors. Neural Networks achieved the highest accuracy, while K-Means clustering revealed distinct emission patterns.